NIRVANA – "NIE Research Vehicle for Atmospheric and Navigation Analysis"

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***Abstract*—NIRVANA is a 1U CubeSat ($10 \times 10 \times 10$ cm, ~ 0.5 kg) built from a 3D-printed PLA structure and commercial off-the-shelf electronics. It carries an ESP32 microcontroller (with Wi-Fi) and an Arduino Uno to control sensors (DHT11 for temperature/humidity, MPU6050 3-axis gyro/accelerometer) and manage data flows. Power is supplied by a 7×7 cm solar panel charging an 18650 Li-ion battery (with BMS) and regulated to 3.3 V/5 V for the electronics. NIRVANA was tested for 24 hours at refrigerator temperatures, where all components remained operational. The system collects > 2000 data points (temperature, humidity, IMU) and transmits them via Wi-Fi to the ThingSpeak IoT platform for open data access. A deployable end-of-life deorbit device (e.g. a drag sail or micro-thruster) is included to ensure atmospheric reentry. NIRVANA aims to provide real-time environmental sensing and situational awareness for border regions at a fraction of the cost of traditional satellites, with rapid deployment and self-deorbit capability.**

***Index Terms*—CubeSat, NIRVANA , IoT, Telemetry, ESP32, ATMega-328p, Earth Observation, Deorbit Mechanism.**

I. Introduction

Earth observation (EO) satellites have long been crucial for applications such as environmental monitoring, climate analysis, and Earth Observation . Large EO platforms like India's RISAT and Cartosat series deliver high-resolution imagery, while European missions such as Sentinel-6 provide ocean topography and atmospheric data through advanced altimeters and thermal sensors. However, these conventional satellites are typically expensive—costing tens of millions of dollars—and require years for development and deployment. Their limited flexibility, slow update rates, and dependence on dedicated ground infrastructure often restrict timely situational awareness in dynamic scenarios like environmental hazards.

The rapid evolution of small satellite technology, particularly CubeSats, presents a disruptive alternative to these traditional architectures. CubeSats are standardized, modular satellites (typically 1U: 10 × 10 × 10 cm, 1 kg) that leverage commercial off-the-shelf (COTS) components. They can be developed within two years at a fraction of the cost of traditional satellites and deployed in large numbers as secondary payloads. For instance, ISRO's PSLV-C37 mission launched 104 satellites in a single flight, demonstrating the scalability and cost-efficiency of CubeSat deployments.

Moreover, CubeSats offer advantages beyond affordability and speed. When equipped with Internet of Things (IoT) capabilities, they can transmit environmental data directly to public cloud platforms in real time, bypassing conventional ground station bottlenecks. Additionally, small satellites typically burn up upon atmospheric re-entry, minimizing the risk of long-term space debris. These features position CubeSats as ideal platforms for distributed environmental sensing and continuous monitoring of inaccessible regions. To address the challenges posed by large satellite limitations, the NIRVANA (Orbital Research Based Low-cost IoT) 1U CubeSat CubeSat was developed as a lightweight, low-cost, and rapidly deployable platform. NIRVANA integrates microcontrollers, sensors, and real-time wireless telemetry to enable autonomous data collection and broadcasting. With a built-in passive deorbit system and support for IoT-based data sharing, NIRVANA demonstrates a sustainable and scalable approach to space-based environmental monitoring and Earth Observation .

II. Related Work

Several research initiatives have investigated compact and low-cost satellite platforms for environmental monitoring and defense-oriented applications. Notable student-driven missions such as ISRO's STUDSAT and Jugnu successfully demonstrated nanosatellite capabilities in Earth observation and atmospheric data acquisition, validating academic participation in space missions. Similarly, NASA's PhoneSat program showcased the feasibility of employing consumer-grade embedded electronics for space applications, highlighting cost efficiency and processing capability in constrained satellite systems.

Recent CubeSat missions, including ESA's CubeMAP and Brown University's EQUiSat, have emphasized educational and open-source CubeSat platforms integrating IoT-enabled telemetry for near-real-time sensor data transmission. Additionally, several studies have explored the use of Wi-Fi based communication links in small satellites, confirming their viability for short-range, low-Earth-orbit operations. However, many existing systems remain limited by monolithic controller architectures, inefficient power-aware data handling, and restricted modularity in sensor integration.

The NIRVANA – Orbital Research Based Low-cost IoT CubeSatellite addresses these limitations by implementing a dual-microcontroller architecture that decouples sensor data acquisition from IoT-based communication, enabling reliable cloud-connected telemetry. Furthermore, NIRVANA incorporates modular sensor interfacing, autonomous power regulation, and a passive drag-sail deorbit mechanism, positioning it as a scalable, sustainable, and mission-adaptive CubeSat

framework for Earth observation and environmental monitoring applications.

A. System Architecture

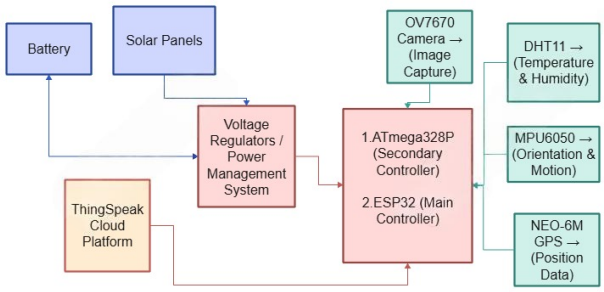

Fig. 1. System architecture of the NIRVANA satellite illustrating communication flow between power, control, and sensor subsystems.

1) Subsystems: The NIRVANA system is based on a 1U CubeSat form factor (10 × 10 × 10 cm) and consists of five major subsystems:

- **Power Subsystem (EPS):** Comprises two 7×7 cm solar panels, a single 18650 Li-ion cell (2500 mAh), and voltage regulators (5 V and 3.3 V). A Battery Management System (BMS) safeguards charging cycles, while a passive drag sail or micro-thruster facilitates post-mission deorbiting.
- **Command and Data Handling (CDH):** Utilizes an ESP32 microcontroller for primary control and communication tasks, and an ATMega-328p (ATmega328P) for sensor data acquisition. The ESP32 also formats and transmits data.
- **Sensor Suite:** Includes a DHT11 for temperature and humidity measurement and an MPU6050 6-DOF IMU for motion tracking.
- **Communications Subsystem:** The onboard ESP32 transmits sensor data via Wi-Fi during ground and laboratory testing to the ThingSpeak IoT platform; in orbital deployments, this link is intended to be replaced by UHF/VHF or SATCOM modules. Future iterations may include LoRa or SATCOM modules for extended range.
- **Structure:** The frame is fabricated using 3D-printed PLA/PETG, chosen for its lightweight and impact resistance, and reinforced with carbon-fiber filaments where necessary.

2) Data Acquisition and Transmission Workflow: NIRVANA operates in a cyclic sensing-transmission loop:

1) The microcontrollers wake from low-power mode.
2) Sensors are read and time-stamped.
3) Data is formatted into JSON packets.
4) The ESP32 transmits data via HTTPS to ThingSpeak.
5) The system returns to sleep mode to conserve power.

This methodology ensures near-real-time environmental monitoring while optimizing for power efficiency and data accessibility.

3) Deorbit Strategy: To comply with orbital debris mitigation guidelines, NIRVANA includes a passive end-of-life deorbit system. A deployable drag sail (or optionally, a micro solid-propellant thruster) is triggered via onboard timing or remote command after mission completion. This increases atmospheric drag, ensuring re-entry within 5 years, addressing a common limitation of existing small satellites.

4) Structural and Environmental Validation: To simulate orbital thermal conditions, the assembled CubeSat was subjected to a 24-hour refrigeration test at 4 °C. All sensors, the ESP32, and ATMega-328p maintained full functionality throughout, confirming basic resilience to low-temperature operational conditions. Structural analysis was performed using SolidWorks, validating mass distribution, mechanical balance, and dimensional tolerances for launch-readiness.

B. Design Methodology

1) Design and Development: The initial phase of the NIRVANA development focused on defining the mission requirements. A detailed Computer-Aided Design (CAD) model of the CubeSat was developed using SolidWorks, as illustrated below.

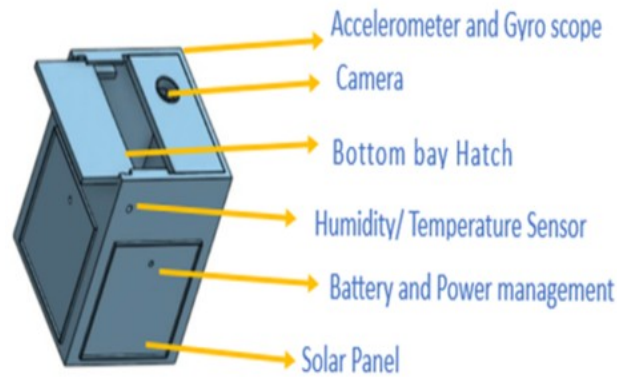

Fig. 2. 3D assembly of the satellite body including the main chassis and bottom bay hatch.

Simulation analyses were conducted to evaluate mechanical strength, thermal behavior, and optimal sensor placement.

2) Reliability Test: Independent validation tests were conducted to assess communication reliability, structural integrity, and the efficiency of the power subsystem.

3) Integration Process: The integrated payload of the NIRVANA CubeSat, as shown in Fig. 3, illustrates the compact and modular arrangement of all onboard subsystems within the 1U CubeSat structure. The payload includes the ESP32 and ATMega-328p microcontrollers, sensor modules, power regulation circuitry, and communication interfaces, organized to ensure efficient space utilization and reliable interconnections. The placement of the battery and power management unit near the structural center improves mechanical balance and stability during launch and operation. Furthermore, the modular stacking configuration adopted in the payload design enables ease of

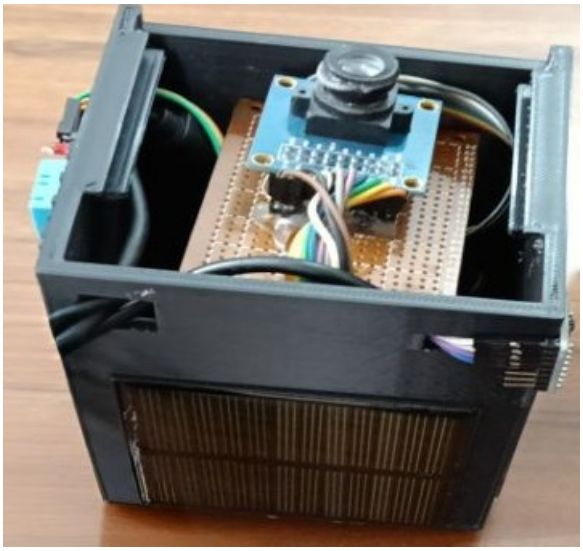

Fig. 3. Payload of the developed NIRVANA satellite showing integrated subsystems.

integration, testing, and future upgrades such as additional sensors or communication modules, without requiring significant structural modifications. This payload configuration validates the feasibility of a low-cost, lightweight, and rapidly deployable CubeSat platform for real-time environmental monitoring and defense-related applications.

III. Results and Discussion

Fig. 4. Live updated temperature measurement from the NIRVANA satellite.

Figure 4 presents the real-time temperature data acquired by the DHT11 sensor onboard the NIRVANA CubeSat. The sensor continuously monitors the ambient temperature surrounding the satellite payload, and the measured values are digitized by the ATMega-328p microcontroller. The processed data is then transmitted to the ESP32 module and uploaded to the ThingSpeak IoT platform for cloud-based visualization. The observed temperature variations validate the reliable operation of the sensing and telemetry system during prolonged operation and demonstrate the suitability of low-cost digital temperature sensors for nanosatellite-based environmental monitoring.

Fig. 5. Live updated humidity measurement from the NIRVANA satellite.

Figure 5 illustrates the real-time relative humidity data captured by the DHT11 sensor integrated within the NIRVANA payload. The sensor measures ambient humidity levels and transmits the digital output to the ATMega-328p microcontroller for initial processing. The processed humidity data is then relayed through the ESP32 module to the ThingSpeak IoT cloud platform, enabling continuous remote monitoring and data visualization. The stable and consistent humidity readings demonstrate the reliability of the sensor integration and confirm the effectiveness of the onboard data acquisition and telemetry system for environmental parameter monitoring.

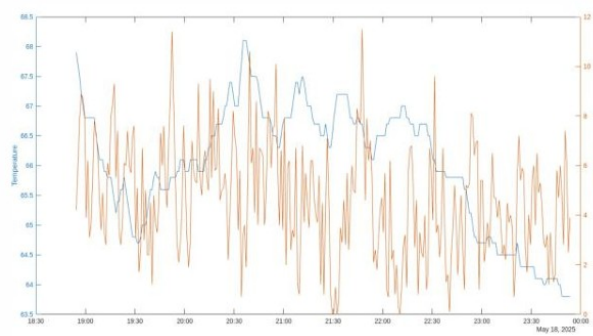

Fig. 6. Real-time airflow speed (ground testing) measurements obtained by the NIRVANA CubeSat using onboard sensors and transmitted to the ThingSpeak IoT platform, demonstrating the satellite's capability for continuous environmental monitoring and predictive data analysis.

Fig. 7. First image captured by the NIRVANA CubeSat using the OV7670 camera module, illustrating the onboard imaging capabilities for real-time visual monitoring and Earth observation.

The first image captured by the NIRVANA CubeSat using the OV7670 camera module. This demonstrates the satellite's onboard imaging capability, enabling real-time visual monitoring and Earth observation for environmental and Earth Observation applications. Images were captured at QVGA resolution and stored locally during ground testing.

Fig. 8. Refrigeration chamber test to assess the performance and resilience of the NIRVANA CubeSat under low-temperature conditions, simulating orbital thermal environments.

Figure 8 illustrates the CubeSat undergoing a 24-hour low-temperature test in a refrigeration chamber (4 °C) to validate structural integrity, sensor operation, and overall system stability under low-temperature operational conditions.

Fig. 9. Satellite orientation monitoring using MPU6050 sensor.

Figure 9 illustrates the NIRVANA CubeSat's implementation of the MPU6050 sensor for precise orientation monitoring. The sensor measures 3-axis angular velocity and linear acceleration, enabling real-time attitude determination, motion tracking, and stabilization analysis, which is critical for maintaining satellite control, payload alignment, and accurate environmental data acquisition during orbital operations.

Future Scope

Future developments of the NIRVANA project will focus on integrating advanced AI-based onboard processing for intelligent event detection and autonomous decision-making. Enhancements in the communication system, such as the use of SDR (Software Defined Radio) and hybrid LoRa–SATCOM links. The inclusion of miniaturized imaging payloads, adaptive solar power management, and low-power deep learning modules will further improve mission endurance and efficiency. Additionally, NIRVANA will be tested for drone-assisted near-space deployment and recovery mechanisms to demonstrate rapid deployment and reusability for short-term Earth Observation or atmospheric missions. The ultimate goal is to scale this system into a small constellation of nanosatellites to provide continuous coverage for scientific research and civilian monitoring applications.

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